

Sviluppi di Taylor di alcune funzioni elementari in $x_0 = 0$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + o(x^4)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + o(x^4)$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + o(x^8) \qquad \sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + o(x^6)$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + o(x^7) \qquad \cosh(x) = 1 + \frac{x^2}{2} + \frac{x^4}{4!} + o(x^5)$$

$$\tan(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^6) \qquad \tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} + o(x^6)$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + o(x^3)$$

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + o(x^8) \qquad \operatorname{arctanh}(x) = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + o(x^8)$$

$$\arcsin(x) = x + \frac{x^3}{6} + \frac{3x^5}{40} + o(x^6)$$